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Assignment 4

1. Retrieve the mission name, rocket name, organization name, launch site name, destination, launch year, and outcome for all space missions without using table aliases. Sort the results in descending order by launch year and ascending order by mission name.

    a. SELECT

        missions.missionname,

        rockets.rocketname,

        organizations.orgname,

        launch_sites.sitename,

        missions.destination,

        missions.launchyear,

        missions.outcome

    FROM missions

    JOIN rockets

        ON missions.rocketid = rockets.rocketid

    JOIN organizations

        ON rockets.orgid = organizations.orgid

    JOIN launch_sites

        ON missions.siteid = launch_sites.siteid

    ORDER BY missions.launchyear DESC, missions.missionname ASC;

    b.

missionname	rocketname	orgname	sitename	destination	launchyear	outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

2. Retrieve the mission name, rocket name, payload name, payload type, and payload mass (in kg) for all payloads launched in 2024 without using table aliases. Sort the results in descending order by payload mass.

    a. SELECT

        missions.missionname,

        rockets.rocketname,

        payloads.payloadname,

        payloads.payloadtype,

        payloads.masskg

    FROM payloads

    JOIN missions

        ON payloads.missionid = missions.missionid

    JOIN rockets

        ON missions.rocketid = rockets.rocketid

    WHERE missions.launchyear = 2024

    ORDER BY payloads.masskg DESC;

    b.

missionname	rocketname	payloadname	payloadtype	masskg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100

3. Retrieve all launch sites along with their matching mission names, rockets, and launch years for missions that took place in 2024 without using table aliases, ensuring that

launch sites with no recorded 2024 missions still appear in the results with null values for the mission fields. Sort the results in alphabetical order by launch site name.

a. SELECT

launch_sites.sitename,

missions.missionname,

rockets.rocketname,

missions.launchyear

FROM launch_sites

LEFT JOIN missions

ON launch_sites.siteid = missions.siteid

AND missions.launchyear = 2024

LEFT JOIN rockets

ON missions.rocketid = rockets.rocketid

ORDER BY launch_sites.sitename ASC;

b.

sitename	missionname	rocketname	launchyear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

4. Write a SQL query using a Common Table Expression (CTE) to calculate the total payload mass (in kg) deployed by each organization across all missions without using

table aliases. Retrieve the organization name, headquarters country, organization type, and total payload mass, and sort the results in descending order by total payload mass.

a.
```
WITH organization_payload_mass AS (
    SELECT
        organizations.orgid,
        COALESCE(SUM(payloads.masskg), 0) AS total_payload_mass
    FROM organizations
    LEFT JOIN rockets
        ON organizations.orgid = rockets.orgid
    LEFT JOIN missions
        ON rockets.rocketid = missions.rocketid
    LEFT JOIN payloads
        ON missions.missionid = payloads.missionid
    GROUP BY organizations.orgid
)
SELECT
    organizations.orgname,
    countries.countryname AS headquarters_country,
    organizations.orgtype,
    organization_payload_mass.total_payload_mass
FROM organizations
JOIN countries
    ON organizations.headquarterscountry = countries.countrycode
```

```
JOIN organization_payload_mass
    ON organizations.orgid = organization_payload_mass.orgid
ORDER BY organization_payload_mass.total_payload_mass DESC;
```

b.

orgname	headquarters_country	orgtype	total_payload_mass
National Aeronautics and Space Administration	United States	Government	25800
Space Exploration Technologies Corp	United States	Private	13692
Japan Aerospace Exploration Agency	Japan	Government	6000
Roscosmos State Corporation	Russia	Government	2450
China National Space Administration	China	Government	1200
Indian Space Research Organisation	India	Government	150
Rocket Lab	New Zealand	Private	100
Arianespace	France	Private	0

5. Write a SQL query using two separate Common Table Expressions (CTEs)—one to calculate total missions by country code and another to calculate total payload mass by country code without using table aliases. Then, combine these CTEs with the countries table to display the country name, total number of missions, and total payload mass (in kg). Sort the results in descending order by total payload mass.

    a.
```
WITH mission_totals_by_country AS (
    SELECT
        launch_sites.countrycode,
        COUNT(missions.missionid) AS total_missions
    FROM launch_sites
    LEFT JOIN missions
        ON launch_sites.siteid = missions.siteid
    GROUP BY launch_sites.countrycode
),
payload_mass_by_country AS (
    SELECT
```

```
    launch_sites.countrycode,
    SUM(payloads.masskg) AS total_payload_mass
  FROM launch_sites
  JOIN missions
    ON launch_sites.siteid = missions.siteid
  JOIN payloads
    ON missions.missionid = payloads.missionid
  GROUP BY launch_sites.countrycode
)
SELECT
  countries.countryname,
  COALESCE(mission_totals_by_country.total_missions, 0) AS total_missions,
  COALESCE(payload_mass_by_country.total_payload_mass, 0) AS
total_payload_mass
FROM countries
LEFT JOIN mission_totals_by_country
  ON countries.countrycode = mission_totals_by_country.countrycode
LEFT JOIN payload_mass_by_country
  ON countries.countrycode = payload_mass_by_country.countrycode
ORDER BY total_payload_mass DESC;
```

b.

countryname	total_missions	total_payload_mass
United States	3	39492
Japan	1	6000
Russia	1	2450
China	1	1200
India	1	150
New Zealand	1	100
France	0	0
United Kingdom	0	0

44	19:01:17	SELECT missions.missionname, rockets.rocketname, organization...	8 row(s) returned	0.000 sec / 0.000 sec
45	19:01:18	SELECT missions.missionname, rockets.rocketname, payloads.pa...	8 row(s) returned	0.016 sec / 0.000 sec
46	19:01:18	SELECT launch_sites.sitename, missions.missionname, rockets.ro...	9 row(s) returned	0.000 sec / 0.000 sec
47	19:01:18	WITH organization_payload_mass AS (SELECT organizations.or...	8 row(s) returned	0.015 sec / 0.000 sec
48	19:01:18	WITH mission_totals_by_country AS (SELECT launch_sites.cou...	8 row(s) returned	0.000 sec / 0.000 sec

